

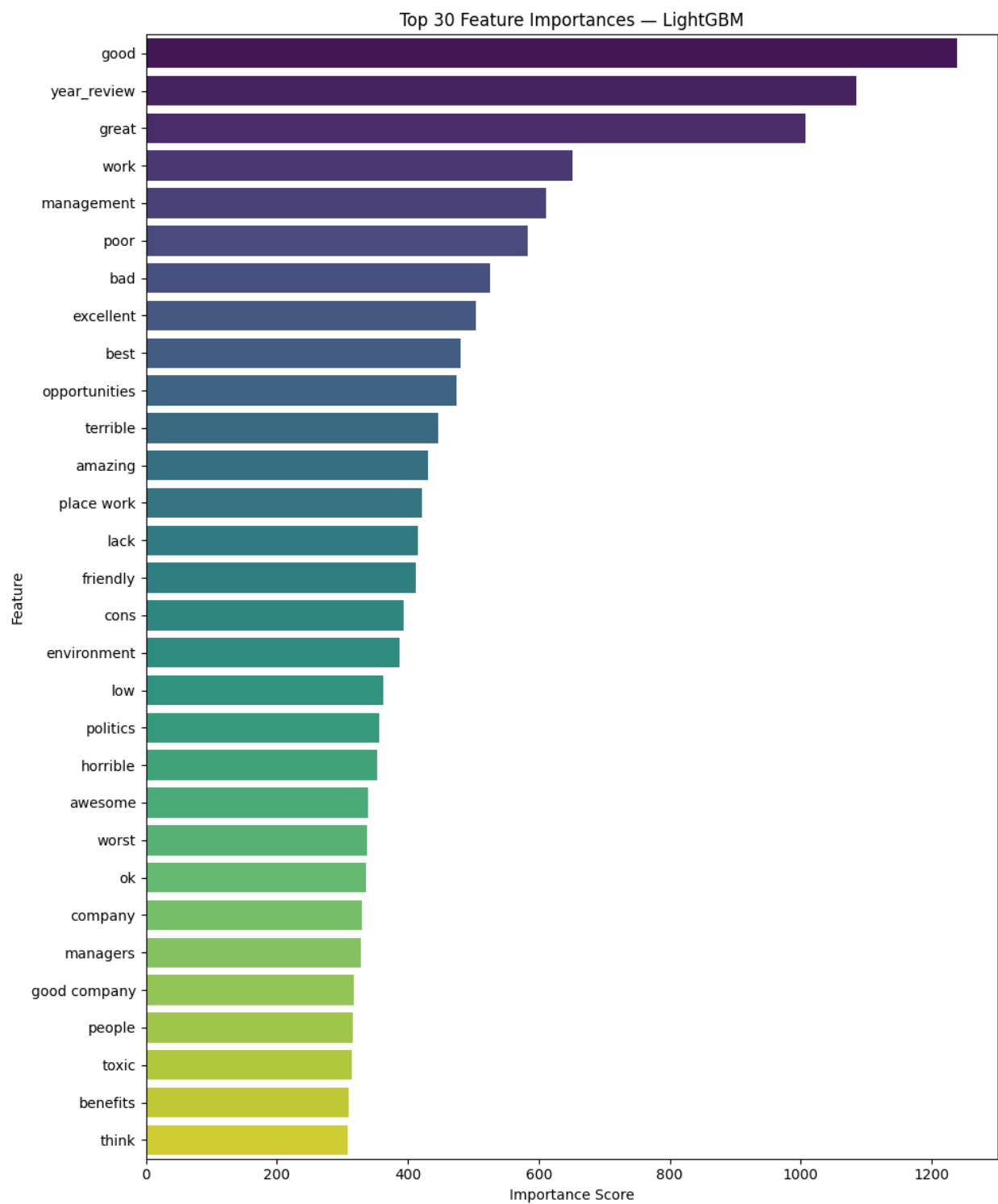
Q1b

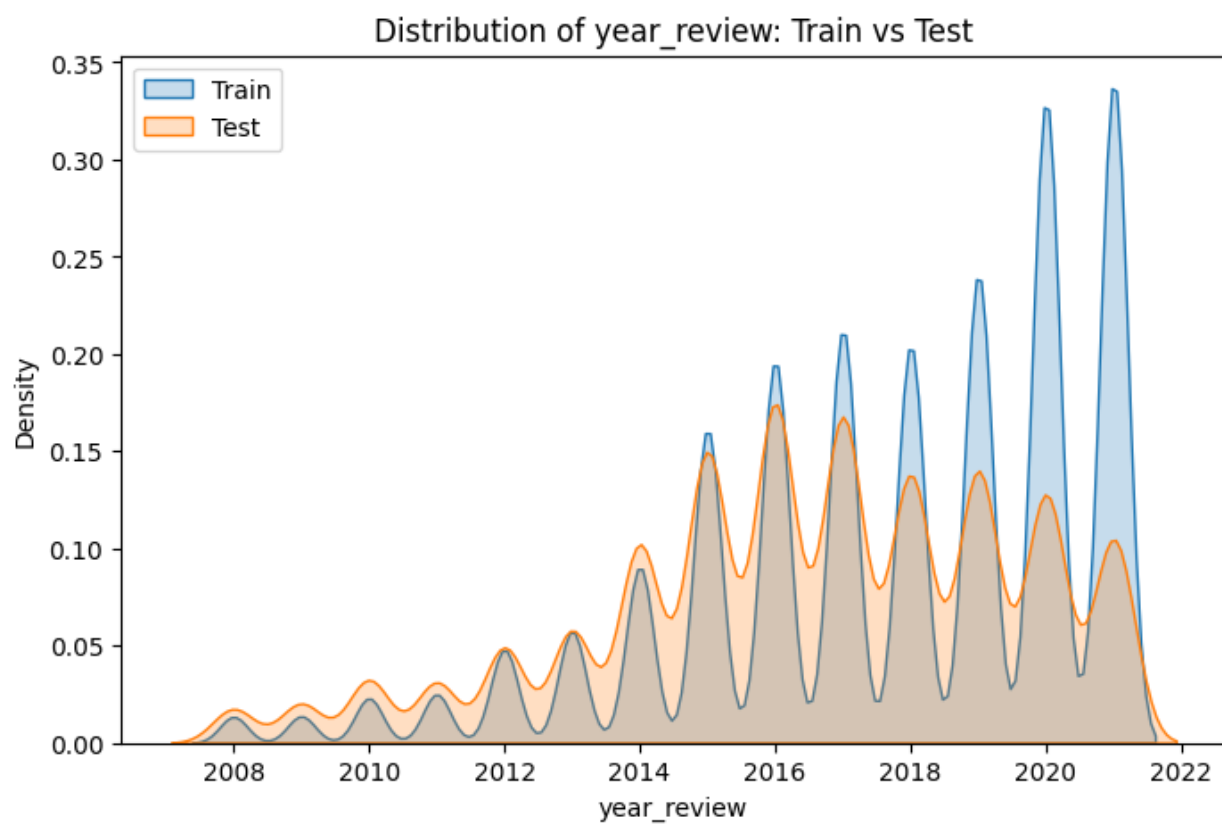
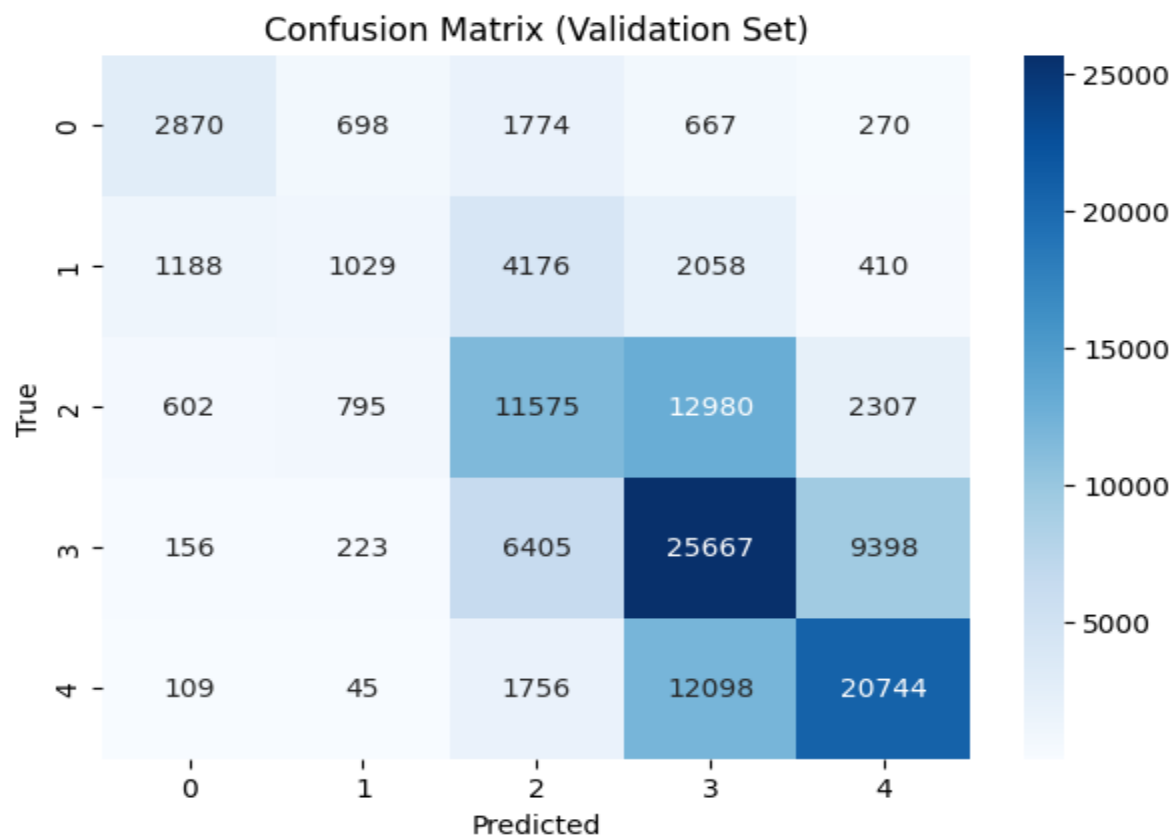
Word Cloud — 1-Star Reviews

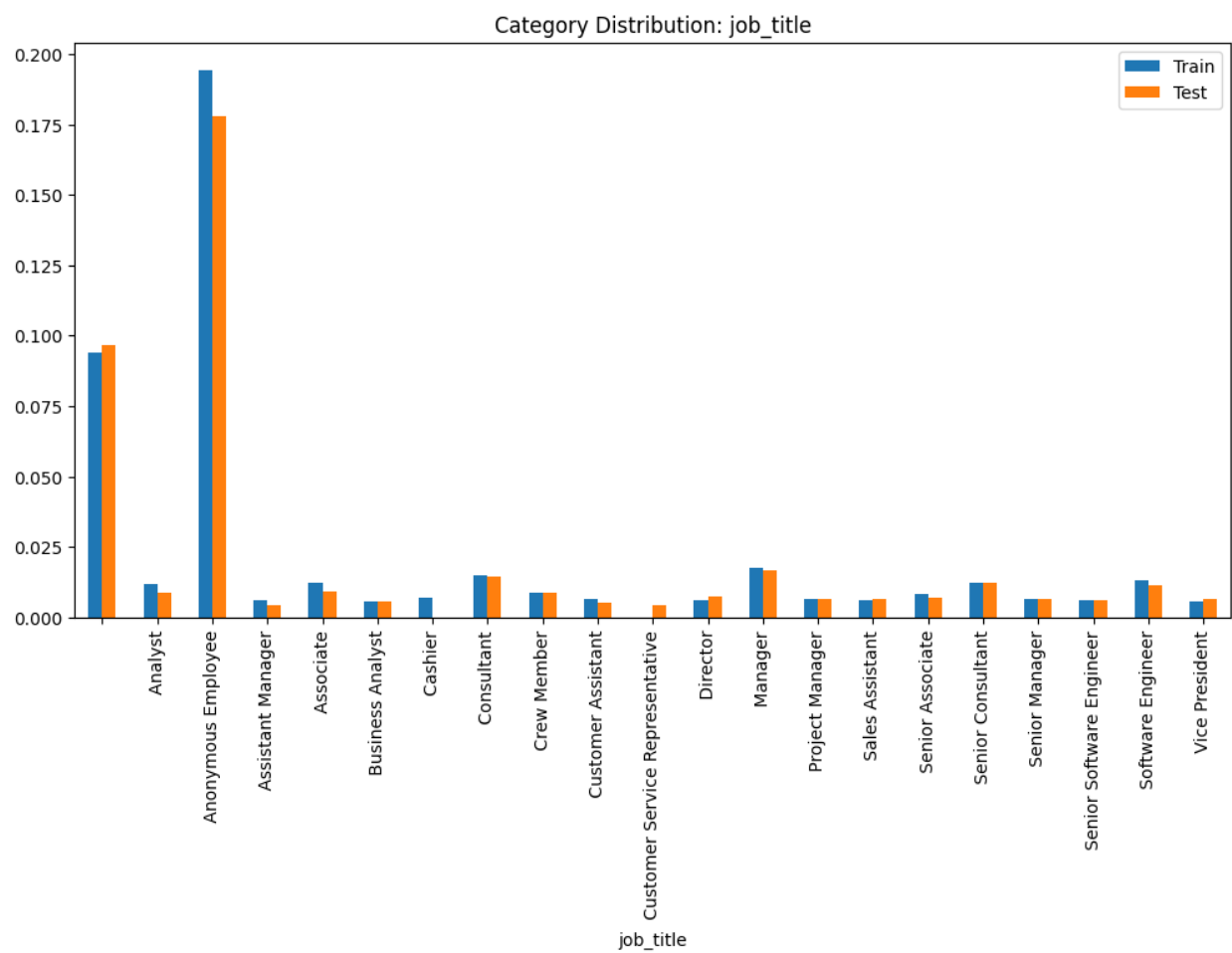
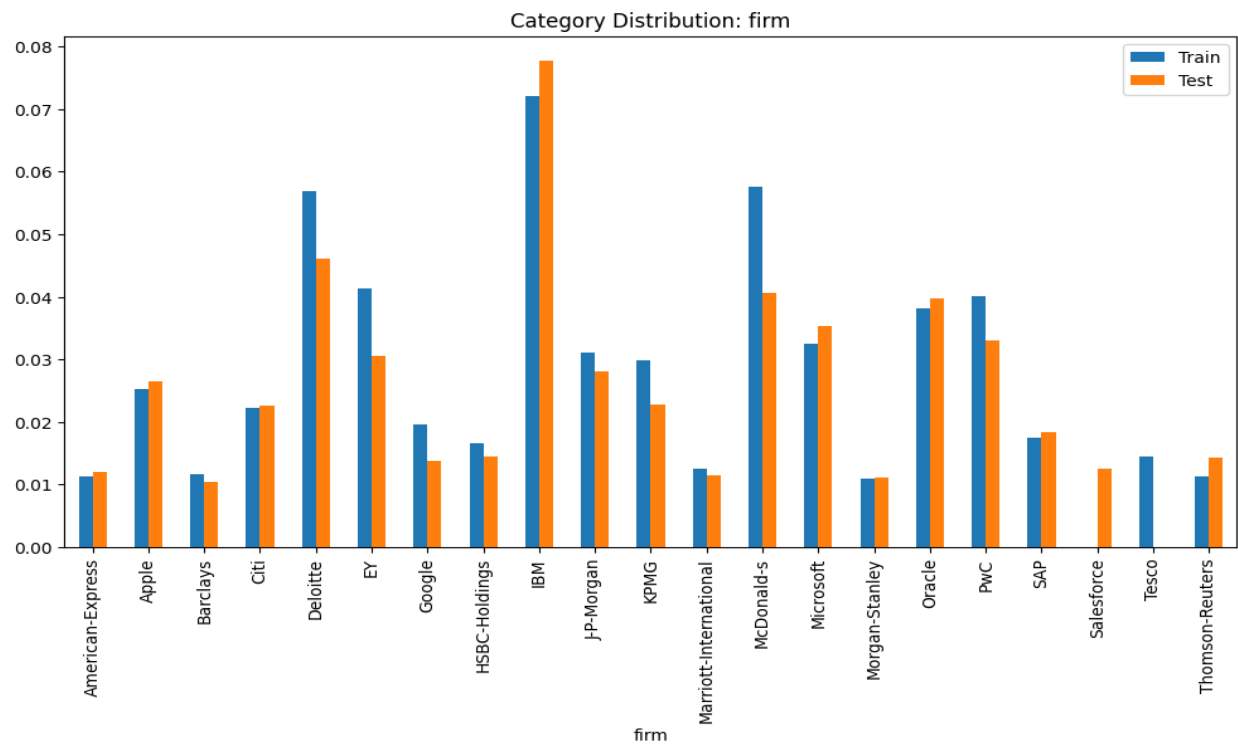


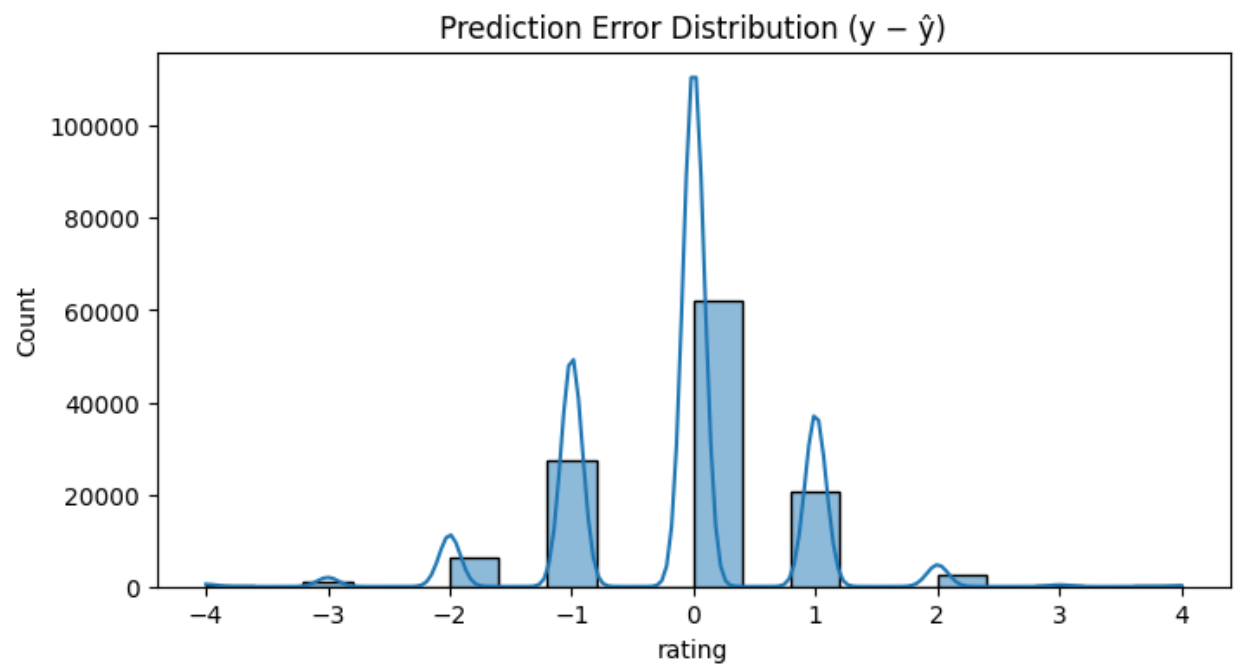
Word Cloud — 5-Star Reviews



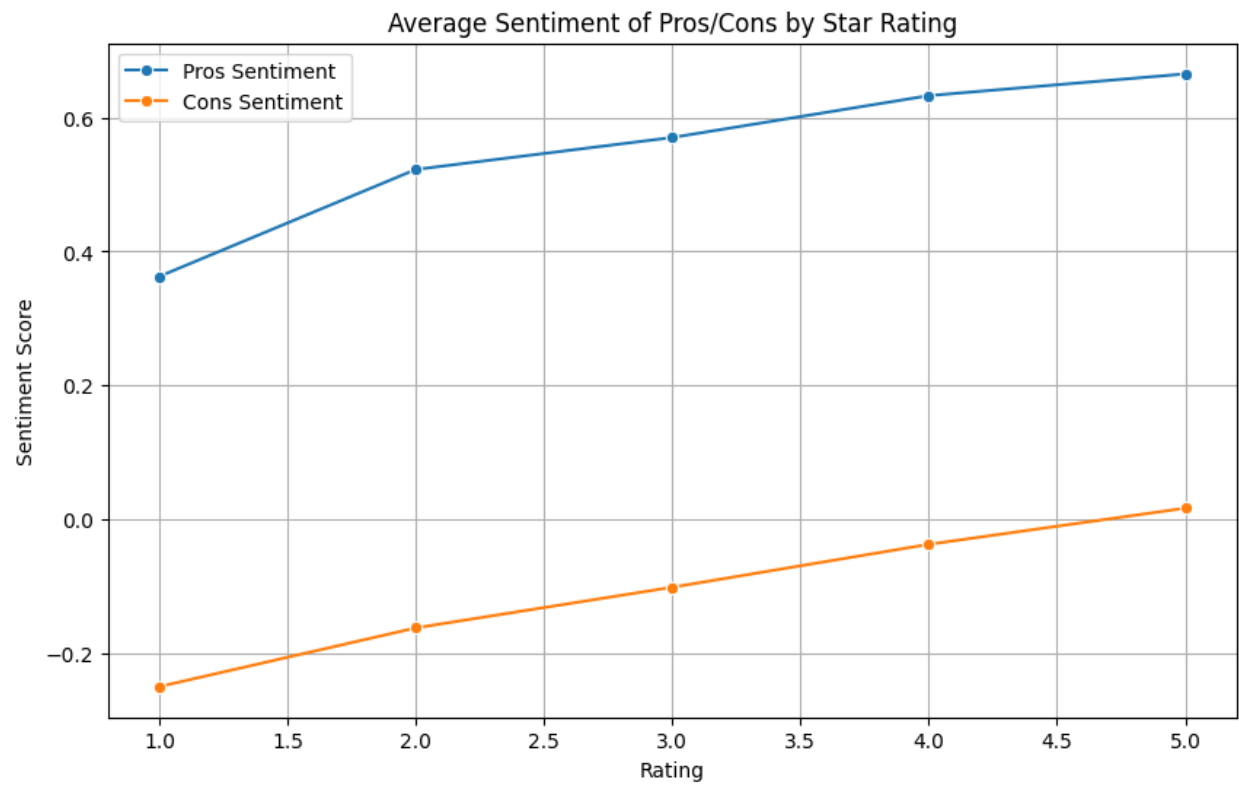
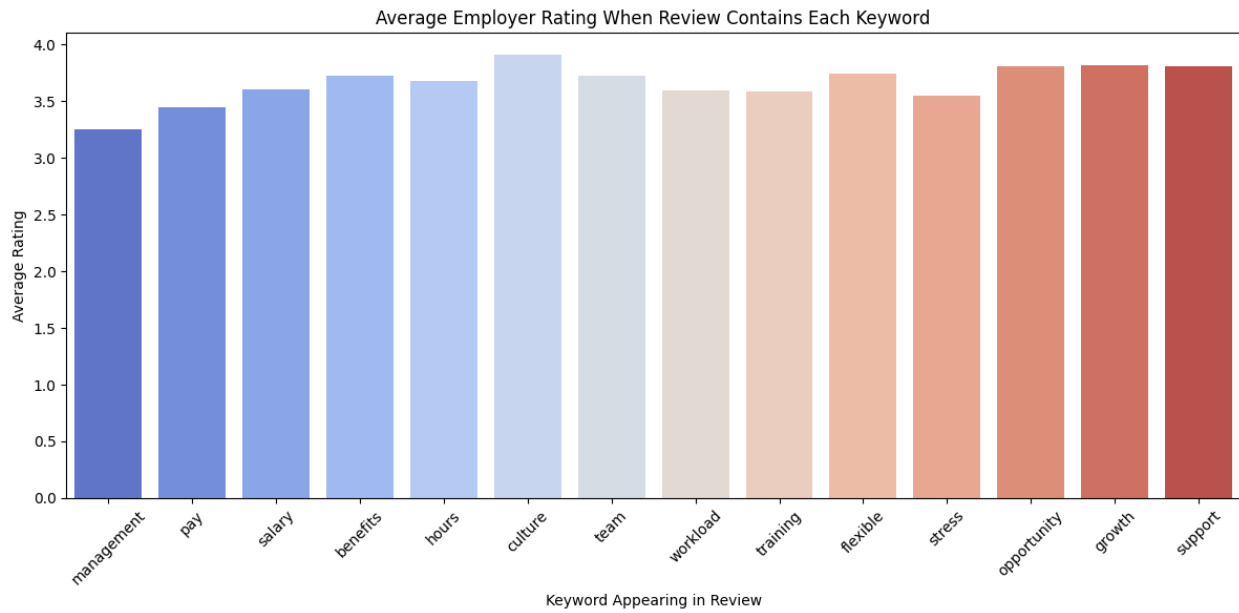








Q1c



Analysis of review text reveals several consistent patterns in what differentiates high-rated companies from low-rated ones. The first key insight is that management quality is the single strongest predictor of employer rating. Reviews containing negative management-related terms (“management”, “leadership”, “communication issues”) are overwhelmingly associated with 1- and 2-star ratings, while high-rated reviews emphasize supportive leadership, collaboration, and a positive team culture. This suggests that firms looking to improve worker satisfaction should prioritize managerial training, communication improvements, and leadership accountability.

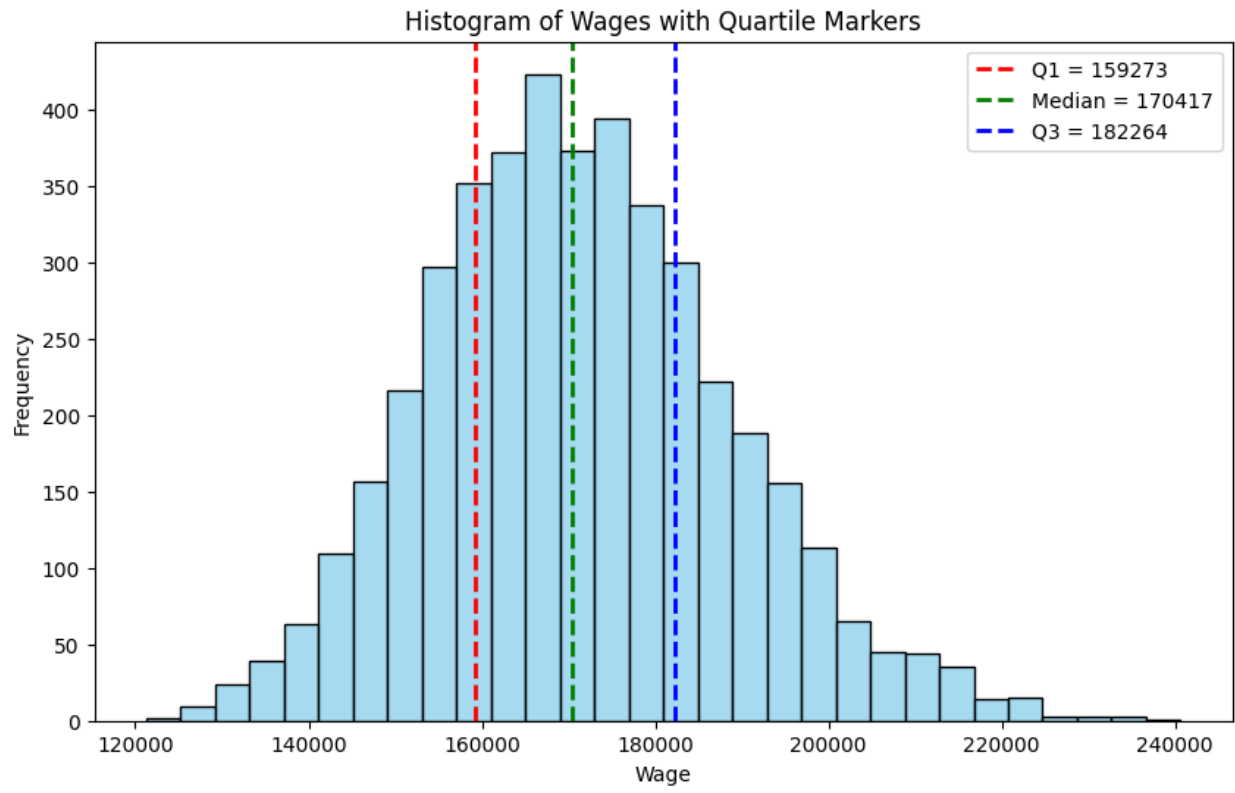
A second insight relates to workload and stress. Mentions of “hours,” “workload,” or “stress” appear disproportionately in low-rated reviews. In contrast, 4- and 5-star reviews commonly highlight “flexibility,” “support,” and “work-life balance.” This points to practical policy levers such as implementing more flexible scheduling, addressing understaffing, and reducing excessive overtime.

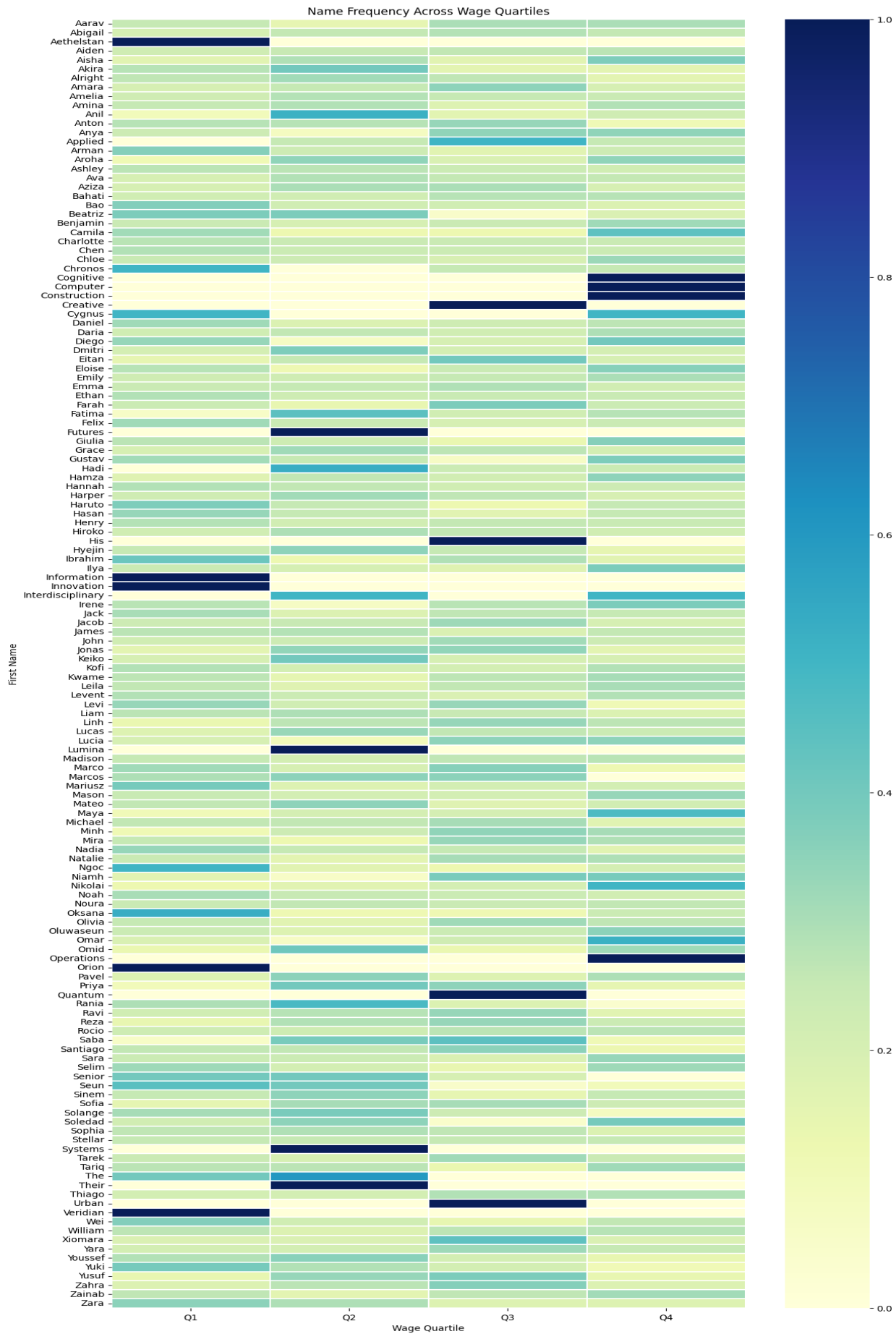
The sentiment analysis further reinforces these findings: sentiment in the “pros” section rises sharply with employer rating, while sentiment in “cons” becomes less negative. This indicates that perceived emotional tone of the workplace matters meaningfully for worker experience.

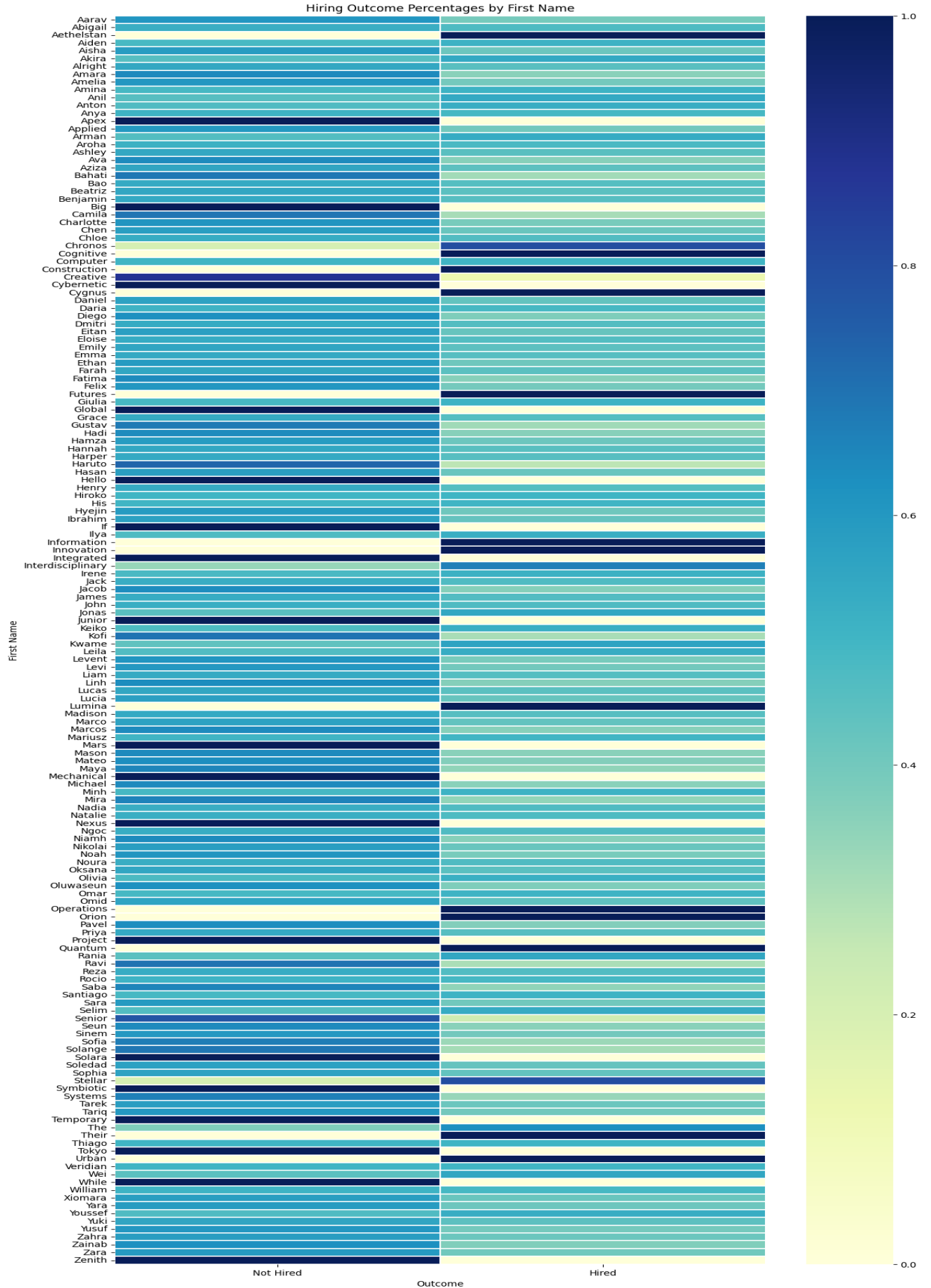
However, while these associations are predictive and informative, they should not be interpreted as causal. Workers self-select into writing reviews, and unobserved factors—such as industry, job type, or personal bias—may confound the relationship between text content and rating. For instance, firms with inherently stressful work environments may attract certain complaints independent of managerial quality. Establishing causality would require experimental or longitudinal data, not observational text.

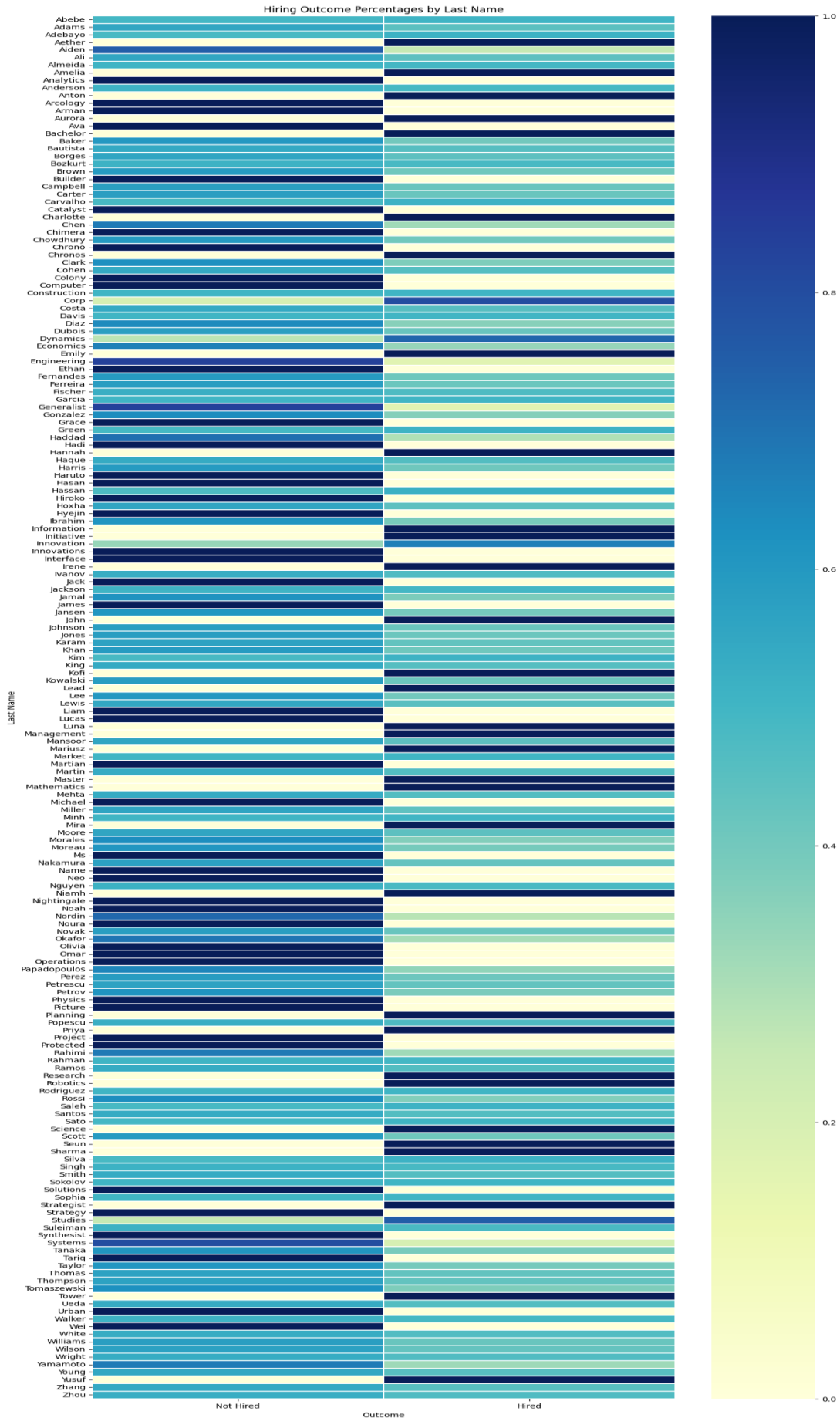
Overall, the data suggests that employers seeking higher satisfaction scores should focus on strengthening management practices, improving communication, and reducing stress through flexible scheduling and manageable workloads.

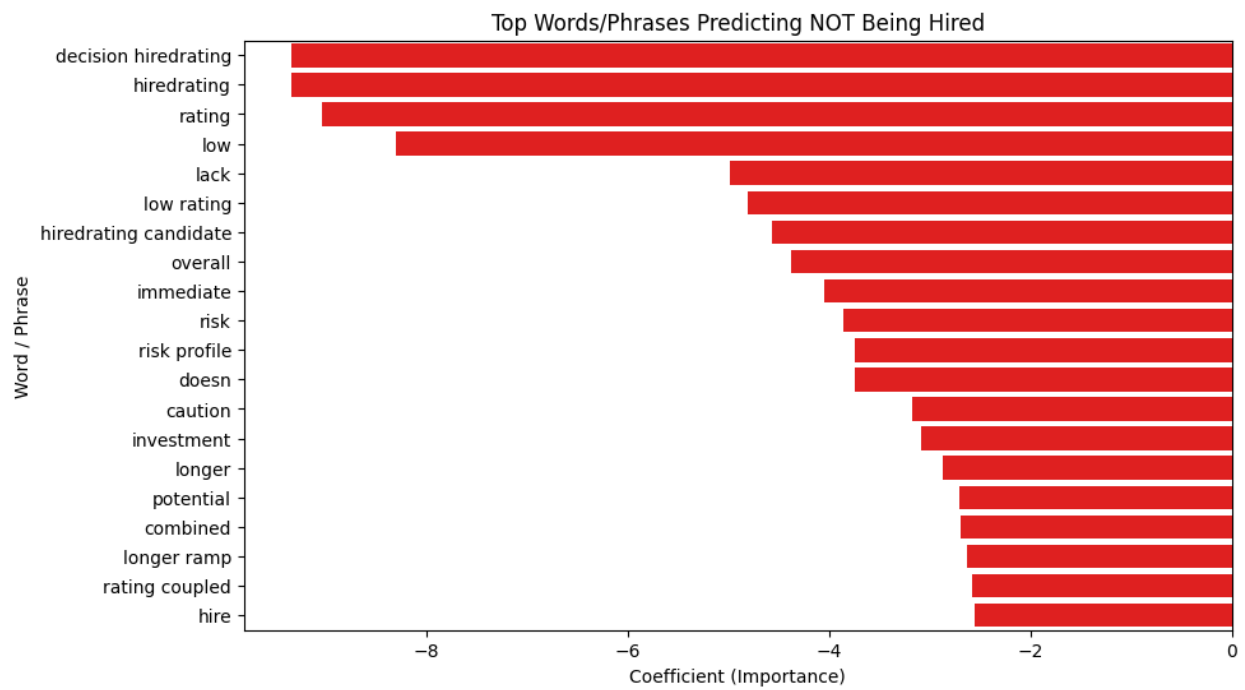
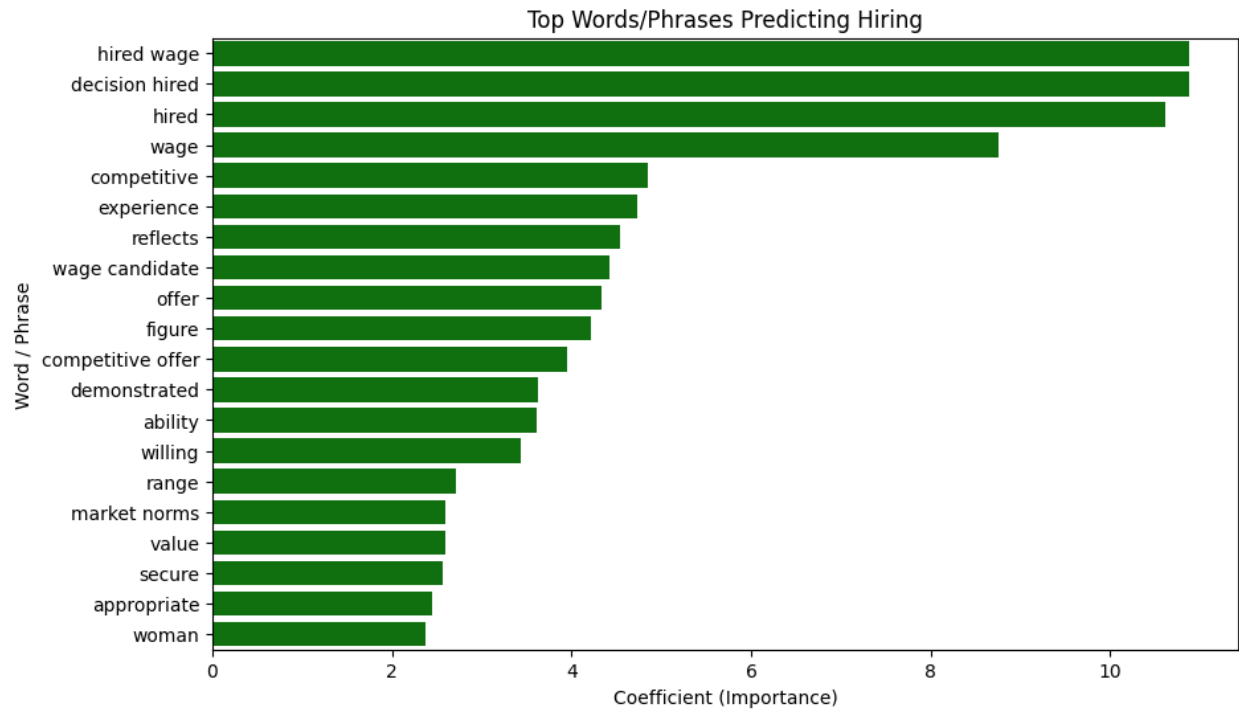
Q2











To evaluate whether discrimination is present in the hiring and wage outcomes, I analyzed patterns in both the hiring decisions (“hired” vs. “not hired”) and the wages offered to successful applicants. Since the dataset does not contain explicit demographic variables, the analysis focused on the information that is available—specifically, the distribution of first and last names across outcomes, and the linguistic content of interview responses. All analysis was normalized to ensure that differences in name frequency did not distort results.

Wage Quartile Analysis

I first divided all hired individuals into wage quartiles and generated a heatmap showing the percentage of each name appearing in each quartile. Normalizing by name was essential because some names appear far more frequently than others. After converting raw counts into within-name proportions, no systematic pattern emerged suggesting that certain names were concentrated disproportionately in either the lowest (Q1) or highest (Q4) wage quartiles. The majority of names displayed relatively balanced distributions across quartiles, and variation that did appear was mostly attributable to names with very small sample sizes. Based on this, there is no strong evidence that wage outcomes differ systematically across names.

Hiring vs. Not Hiring

Next, I compared hired versus not-hired outcomes by computing, for each name, the percentage of applicants who were hired. This normalization avoids misleading conclusions that arise when raw counts are used. The resulting heatmap showed that most names cluster near the overall hiring rate, with no consistent pattern of certain names being disproportionately rejected. A few names appear to have extremely high or low hiring rates, but these correspond almost entirely to cases where the name appears only a handful of times. When restricting attention to names occurring frequently enough to produce stable estimates, hiring rates do not differ meaningfully. Therefore, the hiring data does not provide strong evidence of systematic name-based discrimination.

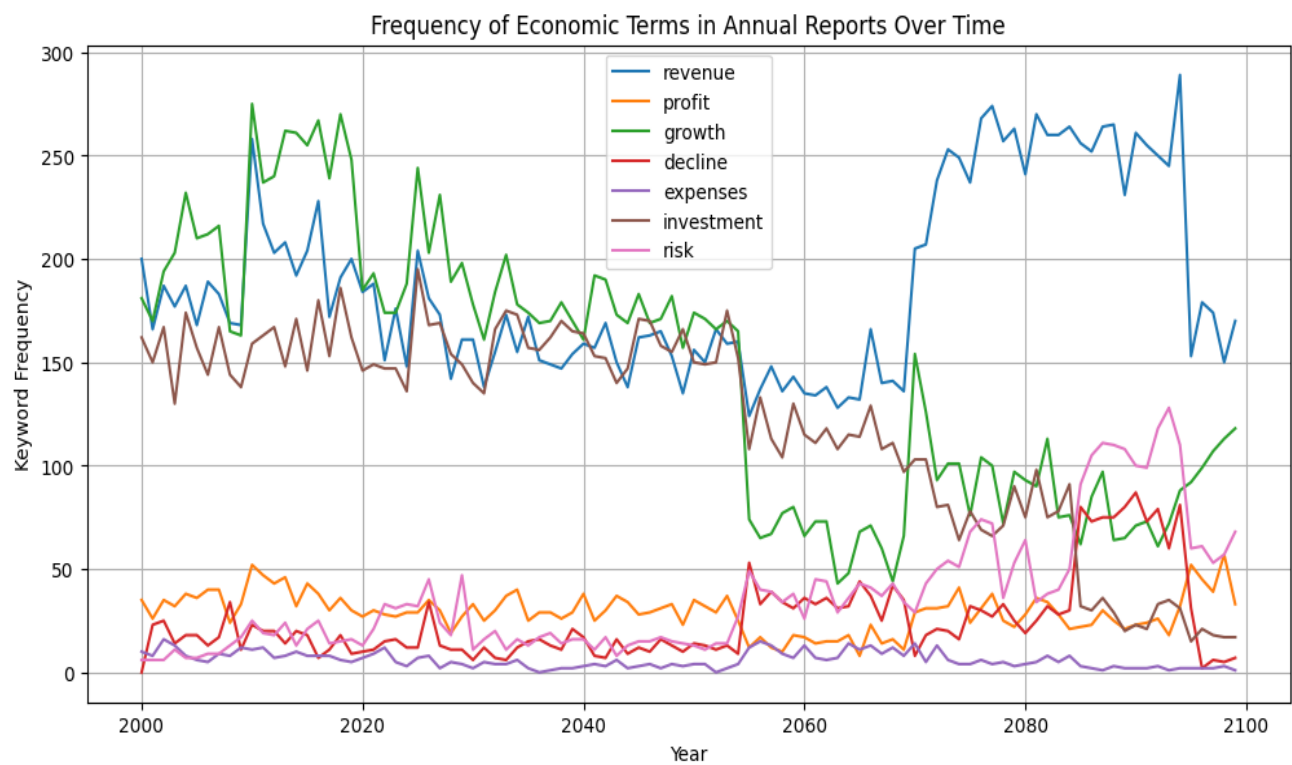
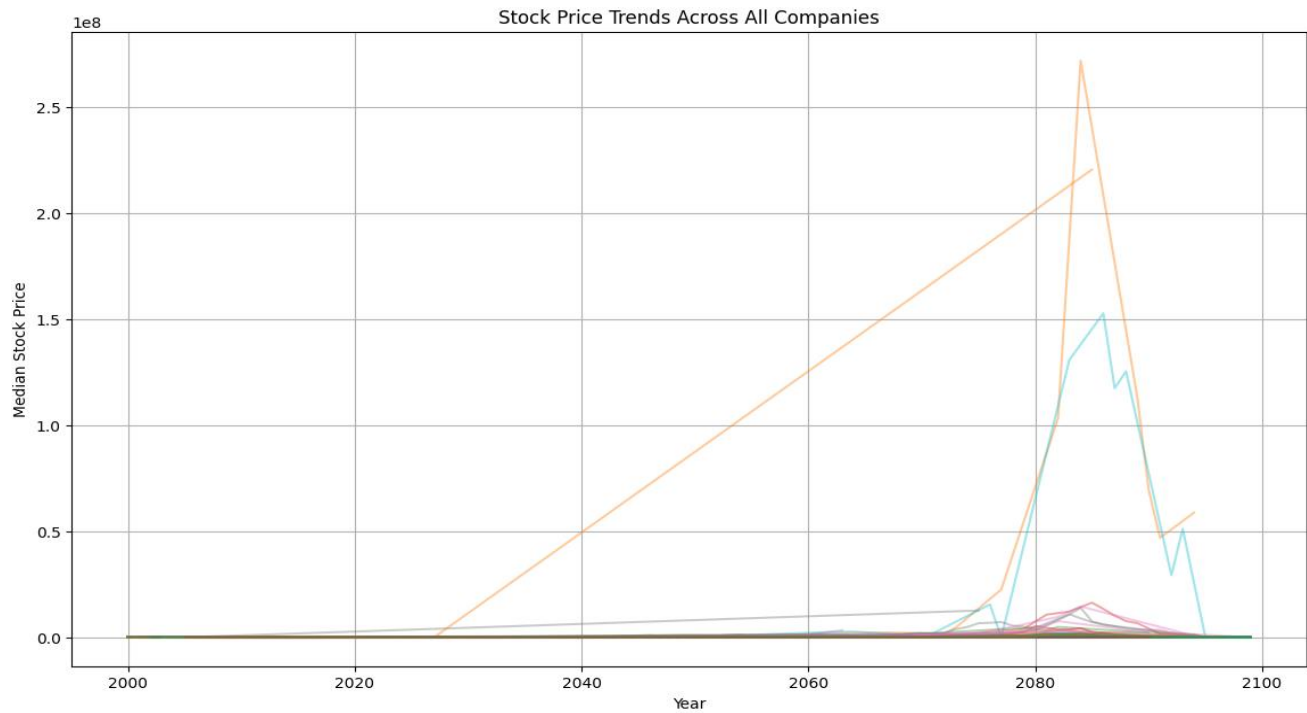
Text-Based Feature Importance

To understand what factors did influence hiring decisions, I trained a text-based classification model using TF-IDF features extracted from applicant discussions and summaries. The most predictive features were job-related terms such as “experience,” “offer,” “competitive,” and explicit mentions of the hiring decision itself. Importantly, neither names nor name-derived tokens appeared among the influential predictors. This suggests that the hiring decisions in the dataset were primarily driven by interview content rather than identity-related cues embedded in names.

Conclusion

Across wage quartile distributions, normalized hiring rates, and model-based text feature importance, there is no strong evidence of discrimination in this dataset. Once name-frequency imbalances are accounted for, neither wages nor hiring decisions appear systematically biased toward or against particular names.

Q3



	year	avg_price	median_price	max_price	min_price
count	100.000000	1.000000e+02	100.000000	1.000000e+02	100.000000
mean	2049.500000	2.494976e+05	11853.590531	1.627609e+07	24.147494
std	29.011492	6.098668e+05	23568.120566	4.545634e+07	48.895717
min	2000.000000	0.000000e+00	0.000000	0.000000e+00	0.000000
25%	2024.750000	1.164841e+02	44.844858	1.394483e+03	0.000000
50%	2049.500000	1.132311e+04	1057.789091	3.410905e+05	0.256851
75%	2074.250000	7.785973e+04	7299.248156	2.356581e+06	25.328835
max	2099.000000	3.516791e+06	118896.213024	2.718207e+08	269.276502

The dataset provides yearly stock prices and annual report text for a large cross-section of companies over roughly a century. To understand the economic conditions during this period, I constructed two indicators: a panel of median stock prices for each company over time, and a text-based index of economic terminology extracted from annual reports. Together, these measures give insight into both market performance and firm-level sentiment.

The first figure plots the median stock price by year across all companies. Several key features emerge. From 2000 through approximately 2050, stock prices remain clustered near zero for nearly all firms, reflecting either a long period of stagnation or limited valuation growth. Beginning around 2060, a subset of firms experiences noticeable upward movement, and by the late 2070s to early 2080s, a dramatic surge occurs. Some companies exhibit extremely rapid price appreciation, peaking around 2085. This period appears to represent a broad economic boom—possibly driven by technological breakthroughs or significant productivity growth.

However, this boom is followed by a sharp decline. Stock prices collapse for many of the previously high-valued firms, and by the early 2090s, prices fall back toward the levels observed earlier in the period. This pattern resembles a speculative bubble, where rapid increases in valuation are followed by equally steep corrections. Even after the decline, a few companies retain moderately elevated prices relative to the earliest years, indicating partial stabilization rather than a full return to pre-boom conditions.

To complement the price data, the second figure measures the frequency of key economic terms—such as revenue, profit, growth, investment, expenses, decline, and risk—in annual reports. These text-based indicators provide qualitative context for the quantitative price

movements. Early in the period (2000–2040), mentions of revenue and growth are relatively high, while terms like risk and decline remain low, suggesting optimism and steady expansion. Between 2040 and 2060, the frequency of positive terms gradually declines, matching the stagnation seen in market prices.

Around 2070, references to revenue, profit, and especially investment rise sharply. This aligns closely with the dramatic stock price surge, suggesting that firms were expanding aggressively, perhaps investing in new technologies or scaling production capabilities. Mentions of risk also rise during this period, indicating heightened uncertainty even as growth accelerates. After the 2085 market peak, references to risk and expenses increase, while mentions of growth fall, consistent with firms managing the aftermath of the bubble's collapse.

Overall, the combined evidence describes an economy that moves from early stagnation to a major innovation-driven boom, followed by a sharp correction and eventual stabilization. The stock market trajectories and textual sentiment from annual reports provide a coherent narrative of speculative expansion, heightened corporate investment, rising uncertainty, and subsequent adjustment.